Conjecture 00000007175: symmetric tridiagonal realizability and the leading-minor chain

Conjecture and reading

Conjecture: "Any real spectrum is realizable by a symmetric tridiagonal, and the positivity of the tridiagonal is the principal-minor chain." We formalize two clauses.

- `Realizable`: for every n and every $\lambda : \{1..n\} \to \mathbb{R}$ (repetitions allowed) there is a real symmetric tridiagonal $n \times n$ matrix A with $\chi_A(X) = \prod_i (X - \lambda_i)$, so multiplicities are retained.
- `MinorCriterion`: for every n and every real symmetric tridiagonal A, A is positive definite iff $\det A_{[k]} > 0$ for all $0 \le k \le n$, where $A_{[k]}$ is the top-left $k \times k$ block.

Tridiagonal means $A_{ij} = 0$ whenever $|i - j| \ge 2$. The Lean statement is the block between STATEMENT BEGIN/END; `Claim := Realizable ∧ MinorCriterion`. Not formalized: any other reading of "chain" (e.g. a three-term recurrence), and any link between the criterion and positivity of the realized eigenvalues.

Proof

Realizability. The diagonal matrix $\mathrm{diag}(\lambda)$ is symmetric, tridiagonal, and $\chi = \prod(X - \lambda_i)$ (Mathlib `Matrix.charpoly_diagonal`). Lean: `C7175.realizable`.

Criterion, $\Rightarrow$. A principal submatrix of a positive definite matrix is positive definite (`Matrix.PosDef.submatrix` along the injective `Fin.castLE`), and positive definite matrices have positive determinant (`Matrix.PosDef.det_pos`). Lean: `C7175.posDef_minors`.

Criterion, $\Leftarrow$ (Sylvester), for every real symmetric matrix, by induction on n. For $n = 0$ the statement is vacuous. For $n + 1$, let B be the leading $n \times n$ block; its leading minors are those of A, so B is positive definite by induction. Reindex $\mathrm{Fin}(n+1) \simeq \mathrm{Fin}\, n \oplus \mathrm{Fin}\, 1$ so $A = \begin{pmatrix} B & b \\ b^T & c \end{pmatrix}$. Since B is invertible, $\det A = \det B \cdot \det S$ with Schur complement $S = c - b^T B^{-1} b$ (`Matrix.det_fromBlocks`$_{11}$). As $\det A > 0$ and $\det B > 0$, $S > 0$. By `Matrix.PosDef.fromBlocks`$_{11}$ (Schur complement criterion), A is positive semidefinite; being of nonzero determinant it is positive definite (`Matrix.PosSemidef.posDef_iff_det_ne_zero`). Lean: `C7175.minors_posDef`. The tridiagonal hypothesis is not needed.

Lean and audit

File `Conjecture7175/Basic.lean` (139 lines). Theorems: `C7175.realizable`, `C7175.minorCriterion`, `C7175.main : Claim`. `#print axioms C7175.main` gives: propext, Classical.choice, Quot.sound. No sorry, no native_decide.